

Introduction

Biofilms are structured microbial societies that are contained in an extracellular polymeric matrix (EPS) from themselves, protecting the bacteria from host immune systems as well as antibacterial treatment. The bubble enables the pathogens to survive for prolonged periods on the surfaces of medical devices, often leading to chronic and persistent infections difficult to treat. Urinary tract infections (UTIs) are becoming increasingly more common due to biofilms growing on urinary catheters.

Fact Category	Statistical Data	Source Context
Annual Medical Visits	8.1 million office visits per year	Most common outpatient infection
Emergency Department Visits	3 million ED encounters annually	Significant acute care burden
Annual Hospitalizations	400,000 hospital admissions yearly	For complicated and severe UTI cases
Women's Lifetime Risk	Over 50% will experience at least one UTI	Higher anatomical vulnerability
Recurrence Rate	27% within 6 months after first episode	High reinfection probability
Primary Causative Agent	75-80% caused by E. coli bacteria	Uropathogenic E. coli dominant
Healthcare-Associated UTIs	68% of hospital UTIs are catheter-related	Most common nosocomial infection
Annual Healthcare Costs	\$3.5-4.8 billion in direct medical costs	Substantial economic impact
CAUTI Rate Range	3.1-7.5 infections per 1,000 catheter-days	Varies by hospital unit type
Mortality from UTI-Related Sepsis	36,000-40,000 deaths annually	From E. coli bloodstream infections

Figure 1. Statistics on UTIs and Their Impact (Aletaha, 2025)

Antimicrobial surfaces are emerging as traditional treatments are becoming less effective. Research on the antibacterial properties show that zinc oxide (ZnO) and titanium dioxide (TiO₂) are strong potential compounds. ZnO demonstrates broad-spectrum antibacterial properties through generating reactive oxygen species (ROS) and the release of Zn²⁺ ions that disrupt bacterial membranes. TiO₂ is widely used for its photocatalytic properties and its ability to generate ROS under light, which can harm bacteria and prevent biofilm formation.

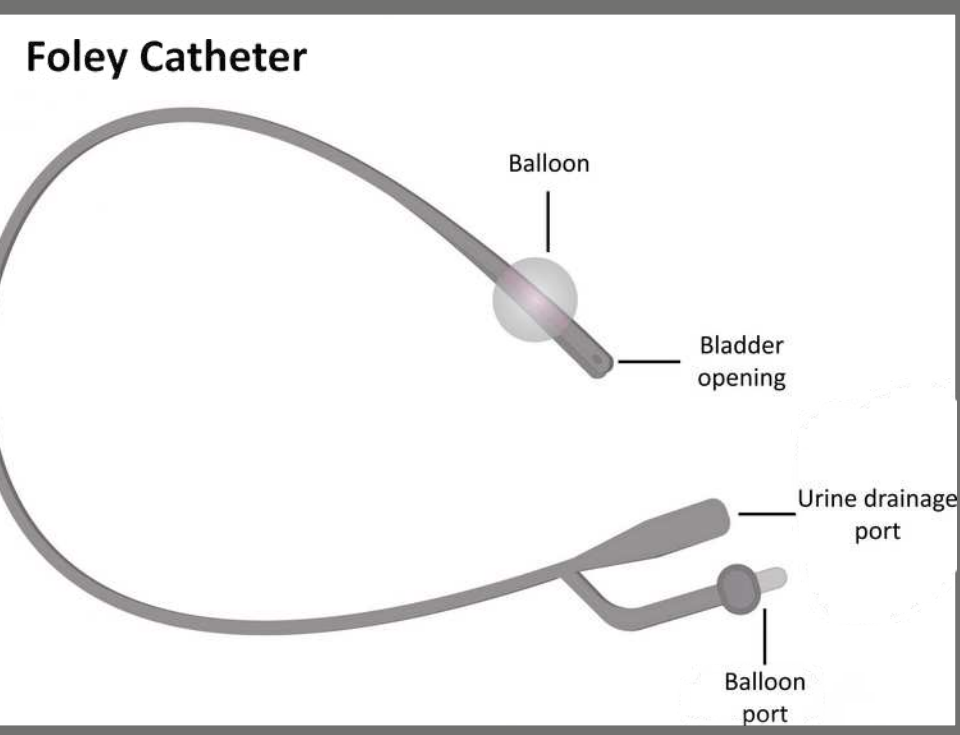


Figure 2. Urinary Catheter Display (Addiss, 2021)

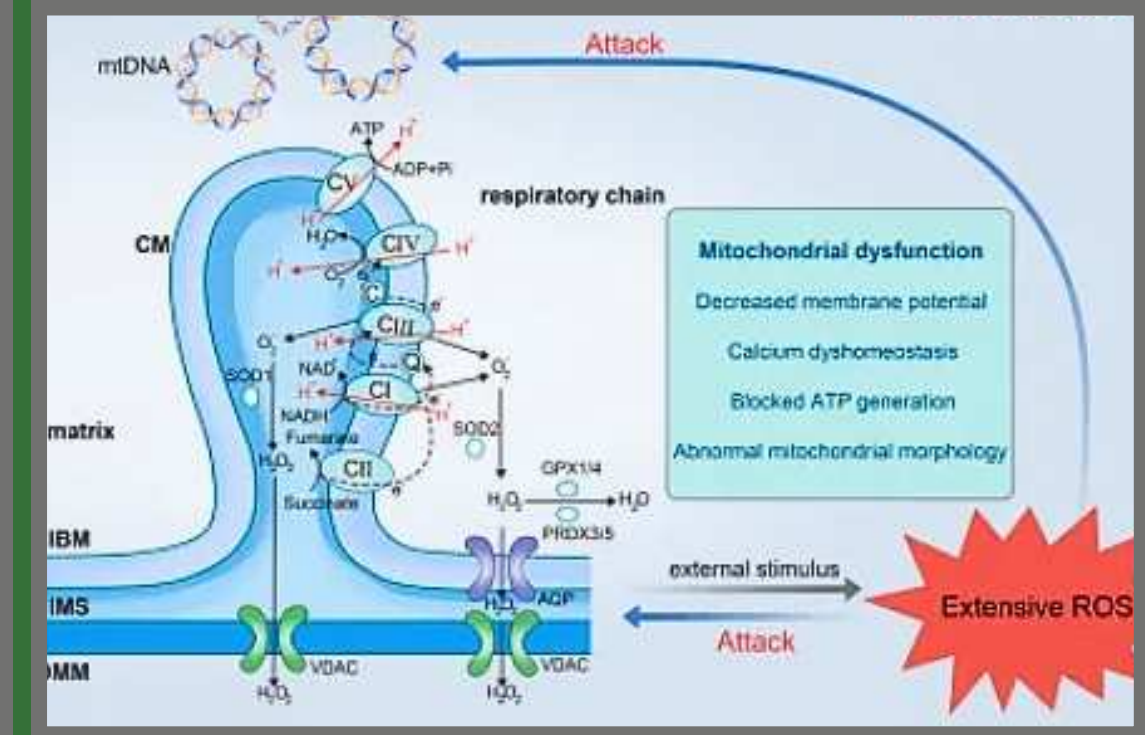


Figure 3. ROS Attacking Cell Membrane (Song et al., 2021)

Purpose

Healthcare-associated infections are still one of the largest challenges to healthcare today, infecting an estimated tens of millions of patients worldwide and extending hospital stays, healthcare expenses, and increasing morbidity. Approximately 65%-80% of infections are biofilm-associated with development on medical devices such as catheters, implants, and prosthetic devices. Between 90% and 100% of patients who undergo long term catheterization develop bacteriuria.

Utilizing a Titanium Dioxide and Zinc Oxide Nanoparticle Coating to Reduce Biofilm Formation on Urinary Catheters

Methods

Polydopamine (PDA) Coating Mixed 50mL deionized water with 100mg of dopamine hydrochloride and 121mg of Trisamino-methane. Then lit sat for 1.5 hours to achieve 50 nanometer thickness so the nanopowders adhere better.



Figure 5. TiO₂ Layer Apperance (Photo taken by Researcher)

Titanium Dioxide (TiO₂) Coating Stirred 40mL deionized water with 10mL 20% ethanol. Added 100mg of Titanium dioxide (TiO₂) and 25mg of Polyvinylpyrrolidone and stirred for two hours. Repeated dip cycle 2 more times to achieve 500 nm thickness.

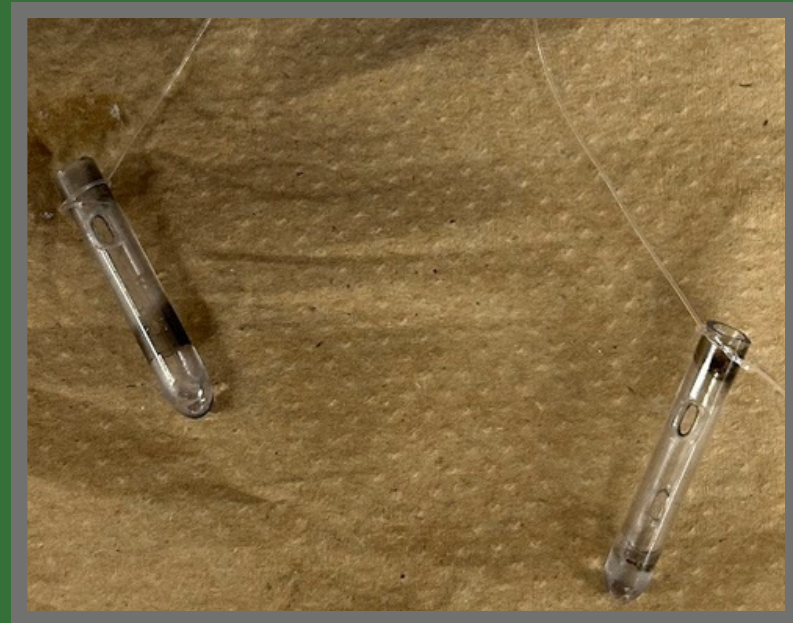


Figure 4. PDA Layer Appearance (Photo taken by Researcher)

Zinc Oxide (ZnO) Coating Stirred 40mL deionized water with 10mL 20% ethanol. Added 250mg of Zinc oxide (ZnO) and 25mg of Polyvinylpyrrolidone and stir for two hour. Repeated dip cycle 4 more times to achieve 3 micrometer thickness.

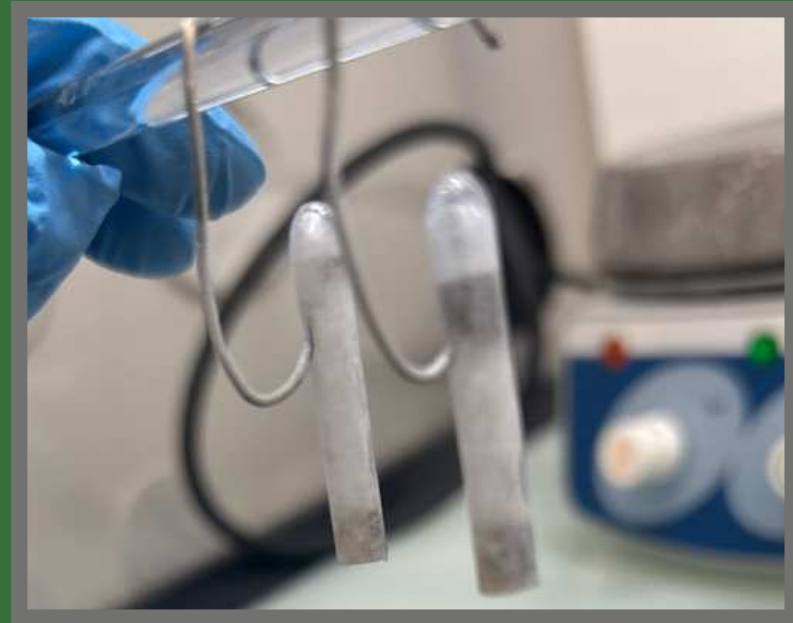


Figure 6. ZnO Layer Appearance (Photo taken by Researcher)

Data Collection

1 Bacteria Preparation



Figure 7. Bacteria Rehydration (Photo taken by Researcher)

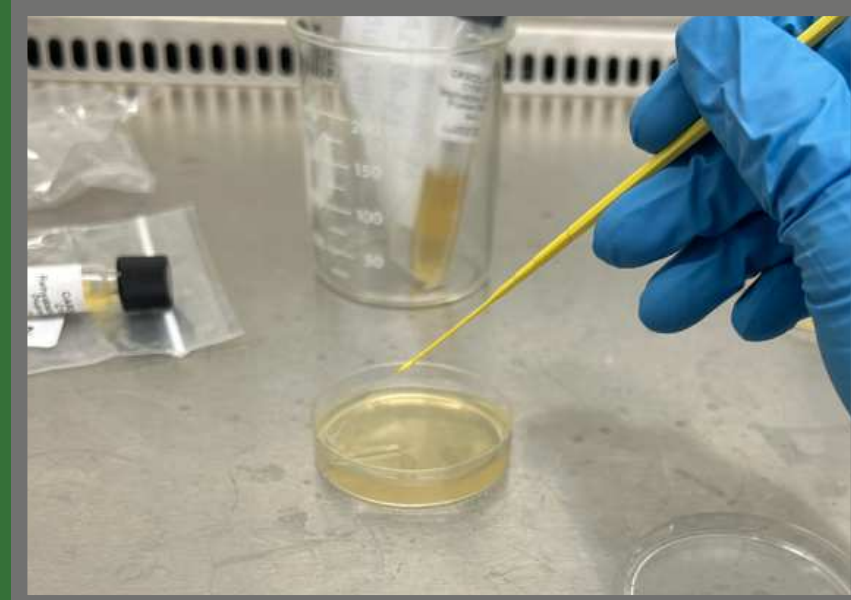


Figure 8. Inoculating Bacteria Onto Agar Plate (Photo taken by Researcher)



Figure 9. Swabbing Cultured Bacteria Onto Catheter (Photo taken by Researcher)



Figure 10. 6-Well Plate incubation (Photo taken by Researcher)



Figure 11. Catheters Submerged In CV (Photo taken by Researcher)



Figure 12. Dried CV On Catheters (Photo taken by Researcher)

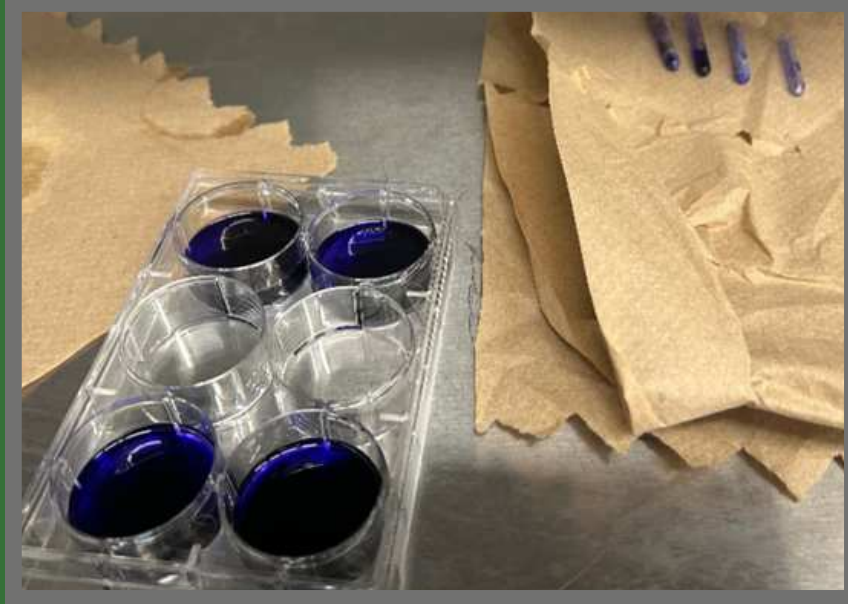


Figure 13. 95% ethanol Solubilizing Solution In Each Well (Photo taken by Researcher)



Figure 14. 20µL Solution Transfer to 96-Well Plate 4x4 area (Photo taken by Researcher)

3 Microplate Reader



Figure 15. SEM Imaging Confirmed Layer Presence(Photo taken by Researcher)

					59
					59
E.coli Uncoated	0.058	0.296	0.235	0.257	59
E.coli Coated	0.137	0.145	0.151	0.144	59
B. subtilis Coated	0.17	0.127	0.182	0.169	59
B. subtilis Uncoated	0.182	0.208	0.224	0.213	59

Figure 16. Absorbance Values Per Well at 595 nanometer absorbance (Table Created By Researcher using Microplate Reader Excel Export)

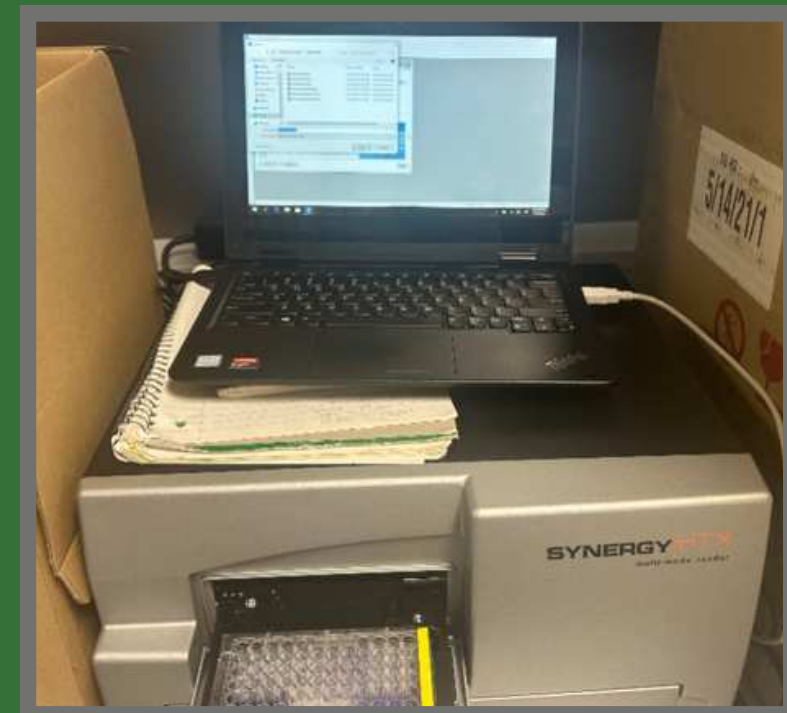


Figure 17. Microplate Reader (Photo taken by Researcher)

Results



Figure 18. E.coli and B.subtilis coated and uncoated absorbance values comparisons (Graph created by Researcher using Google Sheets)

Biofilm Growth Comparison Between Coated and Uncoated (absorbance 595nm)						
	Escherichia coli			Bacillus Subtilis		
	Uncoated (control)	Coated	Percent change from control	Uncoated (control)	Coated	Percent change from control
Average Biofilm Growth	0.2115	0.14425	31.80% Decrease	0.20675	0.162	21.64% Decrease

Figure 19. E.coli and B.subtilis coated and uncoated absorbance values and percent change (Table created by Researcher using Google Sheets)

Conclusion

Biofilm growth decreased on coated surfaces for both species. Escherichia coli showed a ~31.8% reduction and Bacillus subtilis showed a ~21.6% reduction compared to uncoated controls, indicating the coating may inhibit biofilm formation. The lower absorbance values at 595 nm reflects reduced crystal violet retention, supporting decreased biofilm accumulation on coated surfaces. A notable source of uncertainty was variation in the coating itself, as the two coated catheters displayed visible differences in their surface layers. This uneven application could have influenced the extent of biofilm inhibition. Altogether, these findings suggest the coating has antibacterial effects and shows potential for reducing biofilm formation on medical device surfaces.

Extensions

Repeat the study using a more reliable and consistent dip-coating process to ensure uniform application across catheter samples. Test the coated surfaces with replicable human tissue models, such as hydrogels, to assess compatibility with tissue and potential health effects. For results more relevant to real-world medical applications, use bacterial species commonly found on medical devices, such as Staphylococcus epidermidis or Pseudomonas aeruginosa, to better simulate actual biofilm formation. These improvements will help output more accurate results to clarify the coating's antibacterial properties against clinically relevant bacterial species.

References

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